

Michael Pasquale

Nashua, New Hampshire • mrpasquale64@gmail.com • (603) 801-8274 • linkedin.com/in/mrpasquale

SUMMARY

Energetic Senior **Mechanical Engineer** with 10+ years of experience driving cost, efficiency, and **quality** improvements in both **design** and **manufacturing** roles. Ready to provide impactful **engineering** solutions and efficiency improvements in a senior role.

WORK EXPERIENCE

Senior **Mechanical Manufacturing Engineer**

Raytheon Technologies | Andover, MA

February 2020 – Present

- Drove **mechanical engineering** initiatives by designing and optimizing components, tooling, and inspection methods to enhance product reliability, ensure compliance with specifications, and resolve **design** challenges in **collaboration** with vendors and cross-functional teams.
- Engineered custom tooling and **assembly** fixtures for the AN/TPY-2 Radar Defense System components, optimizing **manufacturing** efficiency and enabling a 64% reduction in production time, resulting in \$345,000 in annual cost savings.
- Earned RSTARS award for directing an accelerated **engineering** initiative in **collaboration** with program management, implementing process optimizations leading to critical milestone delivery for \$412M SRP Taiwan Early Warning Radar (EWR) program a year early.
- Designed **assembly** aide fixtures for heavy units, incorporating fully articulating mechanisms that allow assemblers to safely and easily rotate assemblies during production, improving ergonomic safety and reducing **assembly** time by 28%.
- Devised an ergonomic cable management system for a continuity testing station, replacing unsafe overhead setups with adjustable, arm's-reach hangers, eliminating injury risks and reducing cable damage incidents by 60%. Managed **design**, material procurement, and budgeting.
- Facilitated **design** reviews for the tech refresh of the AC/DC radar **assembly**, proposing innovative solutions that reduced **manufacturing** time and costs by 15% while improving ease of testing and **quality assurance** workflows.
- Developed an inspection method using **GD&T** principles to measure shaft run-out, identifying vendor **design** nonconformance that caused defective shafts. Led corrective action with the vendor, eliminating 75% of defects associated with the subassembly that halted production.
- Automated defect trend **analysis** for shop floor systems using Python, integrating SQL data processing to identify and rank trending test failure defects, generating weekly visual reports that reduced manual **analysis** time by 70% and increased first pass yield of AC/DC converters from 48% to 96%.

Mechanical Engineer

Haigh-Farr, Inc | Bedford, NH

March 2018 – February 2020

- Revamped RF antennas for space applications by experimenting with advanced composite **materials**, improving durability by 25% to withstand extreme temperatures and radiation.
- Developed a Python program to convert **SolidWorks** geometry into CNC-compatible G-code, automating a previously manual process and reducing programming time by 80%, enabling faster production cycles and improved precision in tool path generation.
- Achieved over \$2M in contracts by presenting optimized RF antenna designs to aerospace clients, demonstrating resilience to atmospheric reentry and space environment stresses through ANSYS **analysis**.
- Designed multi-use test fixtures, standardizing setup and reducing setup time by 20%, leading to \$6,000 in cost savings.
- Conducted SOW evaluations on new contracts to align existing antenna designs with customer requirements, reducing **design** iterations by 50% and accelerating project timelines.
- Created automated complex curvature generating template utilizing Excel that directly integrated into **SolidWorks**, reducing **design** time of conformal antennas by an average of 3 hours per antenna **design**.

Consultant **Mechanical Engineer** (Contract)

Hartford Steam Boiler | Hartford, CT

July 2017 – January 2018

- Partnered with clients onsite to assess risks and identify gaps in asset failure protection, designing custom sensor solutions leveraging data **analysis** and machine learning to safeguard over \$5M in critical equipment.
- Designed a streamlined, multi-functional sensor for monitoring water, vibration, and temperature, eliminating redundant devices and reducing maintenance costs by 15%, enabling HSB to offer cost-**effective** solutions and attract smaller business clients.
- Created and secured purchase proposals for antenna signal boosters to improve IoT monitoring accuracy by 30% and expand network range, enhancing remote data collection capabilities.
- Adopted **3D** printing for in-house sensor prototyping, reducing lead times by 25% and accelerating iteration cycles for **design** improvements.

Mechanical Engineer (Contract)

Ward Leonard Electric | Thomaston, CT

March 2017 – June 2017

- Collaborated closely with **quality engineering** department in root cause investigations on malfunctioning parts.
- Predicted rotating shaft behaviors, including critical speeds and linear/angular deflection, utilizing ANSYS simulation software, achieving a 95% correlation with physical test results and reducing prototype iteration cycles by 20%.
- Refined motor housing components, optimizing fit to reduce material waste and decrease fabrication costs by \$14,000.

Mechanical Engineer

ARC Technology Solutions, LLC | Nashua, NH

July 2015 – March 2017

- Collaborated with cross-functional teams to **design**, develop, and build automated tooling for jet turbine engine **assembly**, including technical **documentation**, tolerance reviews, and **CAD** simulations to ensure manufacturability and performance.
- Optimized the iterative **design** process for structural components by implementing **SolidWorks** and COMSOL FEA **analysis**, reducing **design** cycle time by 25% while improving simulation accuracy under real-world loading scenarios.
- Designed and implemented a card and label Kanban system for frequently used hardware, streamlining the workflow on the **assembly** floor and decreasing average **assembly** time by 5 hours per build and reducing overstock waste by \$12,000 annually.
- Developed and tested failure strength verification procedures for tooling structural supports, ensuring compliance with customer safety standards and reducing tool-related incidents by 30%.

Junior Mechanical Engineer (Contract)

Aerosat | Amherst, NH

June 2014 – February 2015

- Designed and packaged antenna **communication** systems for commercial airlines, integrating structural supports, cabling, connectors, and electronic components to ensure global operability in harsh environments.
- Developed and implemented a thermal management system using ANSYS, reducing component temperatures by 10 °C to meet performance standards under tight thermal budgets.
- Optimized ACMU system by reducing overall weight by 1.2 kg and cutting production costs by \$15,000 per unit through advanced material selection and Pro/E modeling.

EDUCATION

B.S. Mechanical Engineering

Daniel Webster College • Nashua, NH • 2014

CERTIFICATIONS

Secret Security Clearance

2020

SKILLS

CAD & Simulation: **SolidWorks**, CREO, Windchill, ANSYS, COMSOL

Mathematics & Programming: Matlab, MathCAD, C Basic Programming, VBA, Python

Project & Data Management: MS Project, MS Excel, MS PowerPoint, MS Visio, SAP PRISM, AO Business **Reporting**

Soft Skills: **Collaboration**, Project Management, **Problem**-Solving, **Communication**, Budget Management

Michael Pasquale

- Nashua, NH, US

Contact Information

- b02-fcf-2lz@mail.dice.com
- 6038018274

Work History

Total Work Experience: 10 years

- Senior **Mechanical Engineer** Raytheon
Feb 01, 2020
- **Mechanical Engineer** Haigh-Farr, Inc
Mar 01, 2018
- **Mechanical Engineer** (Contract) Hartford Steam Boiler
Jul 01, 2017
- **Mechanical Engineer** (Contract) Ward Leonard Electric
Mar 01, 2017
- **Mechanical Engineer** ARC Technology Solutions
Jul 01, 2015
- Junior **Mechanical Engineer** (Contract) Aerosat
Jun 01, 2014

Education

- **Bachelors** | Daniel Webster College

Skills

- **solidworks** - 12 years
- **ansys** - 10 years
- **matlab** - 10 years
- **mechanical engineering** - 10 years
- **qa** - 9 years
- **python** - 8 years
- **cost reduction** - 7 years
- **leadership** - 7 years
- **production** - 7 years
- **assembly** - 6 years
- **compliance** - 6 years
- **cross-functional** - 6 years
- **workflow** - 6 years
- **cables** - 5 years
- **corrective and preventive action** - 5 years
- **design review** - 5 years
- **engineering** - 5 years
- **inspection** - 5 years
- **manufacturing** - 5 years
- **manufacturing engineering** - 5 years
- **procurement** - 5 years
- **radar** - 5 years
- **specification** - 5 years

Work Preferences

- Desired Work Settings: Remote or Hybrid
- Likely to Switch: True
- Willing to Relocate: False
- Travel Preference: 25%
- Work Authorization:
 - US
- Work Documents:
 - US Citizen
- Security Clearance: True
- Desired Salary: 120,000+ (USD)
- Third Party: False
- Employment Type:
 - Full-time

Profile Sources

- Dice:
<https://www.dice.com/employer/talent/profile/f98d5b24889733c26bd30b885e4e3a14>